BIOL 250 · Axial SPINAL — PAPER NOTES

QUICK-SCAN INDEX (alphabetical — scan by eye)

All nerve→muscle + all spinal column + reflex key facts. Scan this page first.

TERM	QUICK FACT
Erector spinae	Dorsal rami → extends + laterally flexes vertebral column; intermediate layer; 3 columns (iliocostalis/longissimus/spinalis)
Multifidus	Dorsal rami → extends + rotates column; sacrum + trans. processes → spinous processes 2–4 levels up; deep layer
Semispinalis	Dorsal rami → extension + rotation; cervical + thoracic regions
Splenius capitis/cervicis	Dorsal rami → extends + rotates head; superficial deep back
Axillary n. (C5–C6)	Post. cord brachial plexus → deltoid + teres minor; damage at shoulder dislocation
Common fibular n.	Sciatic branch → tibialis anterior + fibularis longus/brevis + toe extensors; damage → foot drop
Femoral n. (L2–L4)	Lumbar plexus → quadriceps + sartorius + pectineus; damage → weak knee extension
Inferior gluteal n. (L5–S2)	Sacral plexus → gluteus maximus; hip extension
Iliohypogastric n. (L1)	First branch lumbar plexus → ant. abdominal wall + skin
Long thoracic n. (C5–C7)	Roots brachial plexus (no trunk mixing) → serratus anterior; damage → winged scapula
Median n.	Lateral + medial cords → forearm pronators/flexors + thenar; ape hand = thenar atrophy; carpal tunnel
Musculocutaneous n.	Lateral cord → biceps brachii + brachialis + coracobrachialis
Obturator n. (L2–L4)	Lumbar plexus → medial thigh adductors; "obturator foramen"
Phrenic n. (C3–C5)	Cervical plexus → SOLE motor to diaphragm; "C3,4,5 keeps the diaphragm alive"
Pudendal n. (S2–S4)	Sacral plexus → perineum + external sphincters + genitalia
Radial n. (C5–T1)	Post. cord → triceps + all posterior arm/forearm extensors; wrist drop if damaged
Sciatic n. (L4–S3)	Sacral plexus → LARGEST nerve in body; posterior thigh + lower leg/foot
Superior gluteal n. (L4–S1)	Sacral plexus → gluteus medius + minimus + TFL; hip abduction
Tibial n. (L4–S3)	Sciatic branch → posterior leg + plantar foot; plantarflexion (gastrocnemius)
Ulnar n.	Medial cord → intrinsic hand + FCU; "claw hand" (ring + pinky); compressed at medial epicondyle (funny bone)
Annulus fibrosus	Outer fibrocartilage rings of IVD; resists compression/torsion; tears → herniation
Atlas (C1)	No body; no spinous process; ring shape; superior facets support skull
Axis (C2)	Has dens (odontoid process); pivot for head rotation (atlantoaxial joint)
C7 (vertebra prominens)	Longest non-bifid spinous process; most palpable
Cauda equina	Nerve roots/spinal nerves below conus medullaris (L2); looks like horse tail
Conus medullaris	Tapered tip of spinal cord at L1–L2 level
Costal facets	Exclusive to thoracic vertebrae; rib articulation
Denticulate ligaments	Pia mater → dura mater (lateral anchoring of spinal cord)
Epidural space	Between dura + vertebral canal wall; fat + blood vessels (NOT CSF)
Erector spinae	Dorsal rami; intermediate back muscles; extends vertebral column
Filum terminale	Fibrous strand from conus → coccyx; anchors cord
Gray commissure	Where axons cross side to side in spinal cord
Herniated disc	Nucleus pulposus protrudes through annulus fibrosus (posterolaterally)
Lateral horn (spinal)	T1–L2 (sympathetic); S2–S4 (parasympathetic); visceral motor neurons
Ligamentum flavum	Between laminae of adjacent vertebrae; resists flexion; most elastic ligament
Lumbar puncture (spinal tap)	L3–L4 interspace; accesses subarachnoid space → CSF
Nucleus pulposus	Gel core of IVD; remnant of notochord; herniates posterolaterally
Paraplegia	Lower limb motor + sensory loss; below cervical injury
Primary curvatures	Thoracic + sacral; kyphotic; present at birth

Quadriplegia	All 4 limbs; cervical injury
Secondary curvatures	Cervical + lumbar; lordotic; develop postnatally (head lifting + walking)
Subarachnoid space	Between arachnoid + pia; filled with CSF; site of lumbar puncture
Transverse foramina	Exclusive to CERVICAL vertebrae; transmit vertebral arteries + veins
Bell-Magendie law	Dorsal root = sensory (afferent); ventral root = motor (efferent)
Dorsal ramus	Posterior division of spinal nerve → back muscles + posterior trunk skin
Endoneurium	Wraps each individual axon (innermost CT layer)
Epineurium	Wraps entire nerve (outermost CT layer)
Gray ramus communicans	Unmyelinated postganglionic from sympathetic chain → spinal nerve (to skin/sweat/arrector pili)
Perineurium	Wraps fascicle (bundle of axons)
Ventral ramus	Anterior division; forms nerve plexuses; larger
White ramus communicans	Myelinated preganglionic from spinal cord → sympathetic chain
Acquired reflex	Develops with repeated stimulus exposure
Crossed extensor reflex	Paired with withdrawal; contralateral extensors stabilize body
Monosynaptic reflex	Short delay (stretch/patellar); receptor → afferent → motor neuron → muscle
Patellar reflex (knee jerk)	Monosynaptic; L2–L4; femoral nerve; tests L2–L4 integrity
Reflex arc (5 components)	Receptor → afferent fiber → processing center → efferent fiber → effector
Visceral reflexes	Do NOT maintain posture (somatic do); homeostasis
Withdrawal (flexor) reflex	Polysynaptic; pain → interneurons → ipsilateral flexors contract

1. BACK MUSCLES — Axial Musculature

All deep back muscles = Dorsal rami of spinal nerves. Organized deep → superficial.

MUSCLE / GROUP	ORIGIN → INSERTION	ACTION	LAYER / NOTE
Multifidus	Sacrum + transverse processes → spinous processes 2–4 levels up	Extends + rotates vertebral column	Deepest; most important spinal stabilizer
Semispinalis	Transverse processes → spinous processes (cervical + thoracic)	Extension + rotation; cervical region most active	Intermediate-deep
Rotatores (short/long)	Transverse process → spinous process (1–2 levels up)	Rotation (finer)	Deep; segmental
Erector spinae (group)	Sacrum + iliac crest → ribs + vertebrae + skull (variable)	EXTENDS + laterally flexes vertebral column; holds upright posture	Intermediate layer; 3 parallel columns below
— Iliocostalis	Ilium/sacrum → ribs (lateral)	Extends column; lateral column	
— Longissimus	Transverse processes → transverse processes (up) + ribs	Extends + rotates column; middle column	Largest of erector spinae
— Spinalis	Spinous processes → spinous processes (up)	Extends column; medial column	
Splenius capitis	Ligamentum nuchae + spinous C7–T4 → mastoid process + occipital	Extends + rotates head + neck	Superficial deep; neck
Splenius cervicis	Spinous T3–T6 → transverse processes C1–C3	Extends + rotates neck	

2. VERTEBRAL COLUMN

FACT	ANSWER
Total vertebrae	33: 7C + 12T + 5L + 5S (fused) + 3–5 Co (fused)
Intervertebral discs	23
Adult curvatures	4: cervical, thoracic, lumbar, sacral
Primary curvatures (at birth; kyphotic/concave anterior)	Thoracic + Sacral
Secondary curvatures (postnatal; lordotic/convex anterior)	Cervical (head lifting) + Lumbar (walking)
Largest vertebral bodies	Lumbar
Transverse foramina (exclusive to)	Cervical vertebrae (C1–C7); transmit vertebral arteries + veins
Costal facets (rib articulation)	Thoracic vertebrae

Atlas (C1)	NO body, NO spinous process; ring shape; superior facets support skull on occipital condyles
Axis (C2)	HAS dens (odontoid process); atlantoaxial joint = pivot for head rotation
C7	Vertebra prominens: longest non-bifid spinous process; most palpable
Cervical spinous processes	Bifid (C2–C6 typically)
Vertebral foramen size	C = large + triangular; T = circular + small; L = large + triangular
Herniated disc	Nucleus pulposus protrudes POSTEROLATERALLY through annulus fibrosus tears
Ligamentum flavum	Connects LAMINAE; most elastic ligament; resists flexion
Anterior longitudinal lig.	STRONGEST longitudinal lig.; prevents hyperextension
Posterior longitudinal lig.	Inside vertebral canal on posterior body surface; prevents hyperflexion
Transverse lig. of atlas	Holds dens of axis; prevents post. displacement into spinal cord
Ligamentum nuchae	Continuation of supraspinous lig. skull→C7; elastic; holds head upright
True ribs	1–7; direct sternal attachment
False ribs	8–12; 8–10 via costal cartilage; 11–12 = floating (vertebral only)
Thoracic cage bones	Sternum (manubrium + body + xiphoid) + 12 pairs ribs + 12 T vertebrae

2.1 Regional Vertebra Comparison

FEATURE	CERVICAL	THORACIC	LUMBAR
Body	Small; oval	Medium; heart-shaped	Largest; kidney-shaped
Spinous process	Bifid; short (C3–C6)	Long; angled inferiorly	Short; blunt + horizontal
Transverse process	Has transverse foramina (vertebral a.)	Has costal facets	Lacks foramina + facets
Vertebral foramen	Large + triangular	Circular; small	Large + triangular
Unique feature	C1: no body/SP; C2: dens; C7: long non-bifid SP	Rib articulations (costal facets)	Largest bodies; no unique foramina

3. SPINAL CORD — Gross Anatomy & Cross-Section

FACT	ANSWER
Spinal cord extent	Foramen magnum → L1–L2 (conus medullaris)
Conus medullaris	Tapered tip at L1–L2
Cauda equina	Nerve roots below L2 in vertebral canal; "horse's tail"
Filum terminale	Fibrous anchor from conus → coccyx
Cervical enlargement	C4–T1; innervates upper limb
Lumbosacral enlargement	L1–S3; innervates lower limb
Safe lumbar puncture level	L3–L4 (below conus; within cauda equina)
Dorsal root	Sensory (afferent); has DRG (sensory cell bodies)
Ventral root	Motor (efferent)
Spinal nerve type	Mixed (dorsal + ventral root join)
Total spinal nerve pairs	31: 8C + 12T + 5L + 5S + 1Co
H-shaped gray matter — posterior horn	Sensory (interneurons + relay)
H-shaped gray matter — anterior horn	Somatic motor neuron cell bodies (skeletal muscle control)
H-shaped gray matter — lateral horn	Visceral motor (ANS preganglionic): T1–L2 = sympathetic; S2–S4 = parasympathetic
White matter columns (funiculi)	Posterior (dorsal) = ascending sensory; Anterior (ventral) = descending motor; Lateral = both
Axons cross in	Gray commissure (NOT anterior white commissure)
Ascending tracts	SENSORY information TO brain
Descending tracts	MOTOR commands FROM brain
Meninges (outer→inner)	Dura mater → arachnoid mater → pia mater
Epidural space contents	Fat + areolar CT + blood vessels (NOT CSF)
Subarachnoid space contents	CSF; site of lumbar puncture
Denticulate ligaments	Pia mater → dura mater; lateral support of cord
Paraplegia	Lower limbs; spinal injury below cervical
Quadriplegia	All 4 limbs; cervical injury
Hemiplegia	One side of body; brain injury

4. SPINAL NERVES, CT WRAPPINGS & PLEXUSES

4.1 Connective Tissue Wrappings (inside → out)

LAYER	WRAPS	EXAM KEY
Endoneurium	Each individual axon (innermost)	GR CrossSec Label A
Perineurium	Each fascicle (bundle of axons)	GR CrossSec Label D
Epineurium	Entire nerve (outermost)	GR CrossSec Label C or G

Rami: Dorsal ramus → back muscles + posterior trunk skin. Ventral ramus → body wall + limbs (forms plexuses). White rami = myelinated preganglionic fibers TO sympathetic chain. Gray rami = unmyelinated postganglionic FROM chain.

4.2 Nerve Plexuses — Origin, Key Nerves & Muscles

PLEXUS	SEGMENTS	NERVE	MUSCLES / REGION	INJURY CLUE
Cervical	C1–C4	Phrenic (C3–C5)	DIAPHRAGM (sole motor supply)	C3–C5 injury → respiratory failure
Cervical	C1–C4	Ansa cervicalis (C1–C3)	Infrahyoid mm. (sternohyoid, sternothyroid, omohyoid)	
Cervical	C1–C4	Lesser occipital, great auricular	Posterior scalp + ear skin	
Brachial	C5–T1	Musculocutaneous (lat. cord)	Biceps + brachialis + coracobrachialis	Weak flexion/supination
Brachial	C5–T1	Axillary (post. cord)	Deltoid + teres minor	Shoulder dislocation → arm abduction lost
Brachial	C5–T1	Radial (post. cord)	Triceps + ALL posterior arm/forearm extensors	Wrist drop (humeral fracture)
Brachial	C5–T1	Median (lat.+med. cord)	Forearm pronators/flexors + thenar mm.	Ape hand; carpal tunnel; can't oppose thumb
Brachial	C5–T1	Ulnar (med. cord)	Intrinsic hand mm. + FCU	Claw hand (ring+pinky); medial epicondyle compression
Brachial	C5–T1	Long thoracic (roots C5–C7)	Serratus anterior	Winged scapula; NO trunk mixing
Brachial	C5–T1	Thoracodorsal (post. cord)	Latissimus dorsi	
Brachial	C5–T1	Dorsal scapular (root C5)	Rhomboids + levator scapulae	ONLY brachial nerve without trunk mixing
Lumbar	L1–L4	Femoral (L2–L4)	Quadriceps + sartorius + pectineus	Weak knee extension; ↓ patellar reflex
Lumbar	L1–L4	Obturator (L2–L4)	Medial thigh adductors	
Lumbar	L1–L4	Iliohypogastric (L1 — 1st branch)	Anterior abdominal wall	
Lumbar	L1–L4	Lateral femoral cutaneous (L2–L3)	Lateral thigh SKIN ONLY (sensory)	
Sacral	L4–S4	Sciatic (L4–S3)	Hamstrings (+ branches below knee)	LARGEST nerve in body; posterior thigh
Sacral	L4–S4	Tibial (L4–S3)	Posterior leg (gastrocnemius) + plantar foot	Plantarflexion; plantar foot sensation
Sacral	L4–S4	Common fibular (L4–S2)	Tibialis anterior + fibularis mm. + toe extensors	Foot drop (fibular neck fracture)
Sacral	L4–S4	Superior gluteal (L4–S1)	Gluteus medius + minimus + TFL	Hip abduction; Trendelenburg gait if damaged
Sacral	L4–S4	Inferior gluteal (L5–S2)	Gluteus MAXIMUS	Hip extension; climbing stairs
Sacral	L4–S4	Pudendal (S2–S4)	Perineum + ext. sphincters + genitalia	

4.3 Brachial Plexus Organization

Order: Roots (C5–T1) → Trunks (sup/mid/inf) → Divisions (ant+post each) → Cords (lateral/medial/posterior relative to axillary a.) → Terminal nerves. Median n. = ONLY nerve from BOTH lateral + medial cords. Cords named relative to AXILLARY ARTERY.

4.4 Reflexes

REFLEX	TYPE	ARC	NOTE
Stretch (patellar/knee jerk)	Somatic; monosynaptic	Muscle spindle → afferent → anterior horn motor neuron → muscle	SHORTEST delay; L2–L4; automatic muscle length regulation
Withdrawal (flexor)	Somatic; polysynaptic	Nociceptor → interneurons → ipsilateral flexors contract	Protective; paired with crossed extensor
Crossed extensor	Somatic; polysynaptic	Contralateral extensors stabilize body while ipsilateral withdraws	Occurs simultaneously with withdrawal
Autonomic (visceral)	Visceral; polysynaptic	Interoceptors → ANS → smooth/cardiac muscle or glands	Homeostasis; does NOT maintain posture

GR T/F traps: Monosynaptic reflex has LONG delay = FALSE (shortest). Visceral reflexes maintain upright posture = FALSE (somatic do). Patellar = knee jerk = TRUE. Acquired reflexes develop with REPEATED exposure = TRUE.

RAPID-FIRE SPINAL CHEAT SHEET

#	QUESTION	ANSWER
1	Lig. connecting laminae?	Ligamentum flavum
2	Primary curves (at birth)?	Thoracic + Sacral (kyphotic)
3	Transverse foramina exclusive to?	Cervical vertebrae
4	C1 name + unique feature?	Atlas — NO body, NO spinous process
5	C2 name + unique feature?	Axis — HAS dens (odontoid); pivot rotation
6	Spinal cord ends at?	L1–L2 (conus medullaris)
7	Safe LP level?	L3–L4 (below conus)
8	Epidural space contains?	Fat + areolar CT + blood vessels (NOT CSF)
9	Subarachnoid space contains?	CSF
10	Posterior horn = ?	Sensory relay
11	Anterior horn = ?	Somatic motor neuron cell bodies
12	Lateral horn = ?	Visceral motor (ANS): T1–L2 symp; S2–S4 para
13	Axons cross in?	Gray commissure
14	Endoneurium wraps?	Each individual axon
15	Perineurium wraps?	Fascicle
16	Epineurium wraps?	Entire nerve
17	White rami = ?	Myelinated preganglionic TO chain
18	Gray rami = ?	Unmyelinated postganglionic FROM chain
19	Brachial plexus order?	Roots→Trunks→Divisions→Cords→Peripheral nerves
20	Cords named relative to?	Axillary artery
21	Median n. from which cords?	BOTH lateral + medial cords
22	Only brachial n. without trunk mixing?	Dorsal scapular n.
23	Largest nerve in body?	Sciatic (L4–S3)
24	Phrenic n. origin?	C3–C5
25	Gluteus maximus innervation?	Inferior gluteal n.
26	Hip abductors (glut med/min)?	Superior gluteal n.
27	Medial thigh (adductors)?	Obturator n.
28	Foot drop nerve?	Common fibular n.
29	Wrist drop nerve?	Radial n.
30	Monosynaptic reflex delay?	SHORT (not long)

TB HOT SPOTS — SPINAL / VERTEBRAL

Facts from the Stuvia TB that need quick reference — not fully covered above.

SPINAL / VERTEBRAL

TERM / CONCEPT	KEY FACT (answer the TB is looking for)
Rib cage protects	Heart, lungs, liver, spleen, kidneys — ALL organs (D: all of the above)
Primary spinal curves	Thoracic + sacral (kyphotic); accommodate thoracic + abdominopelvic VISCERA
Cervical curve formation	Develops when INFANT LIFTS HEAD; secondary/compensatory curve
Lumbar curve formation	Develops when infant begins to walk; secondary/compensatory
Vertebral body function	Transfers WEIGHT along long axis of vertebral column
Spinous process	Most easily felt on DORSUM (posterior palpation through skin)
Lamina	Cut during spinal surgery (laminectomy) to access cord
Atlas (C1) posterior tubercle	Atlas has POSTERIOR TUBERCLE instead of spinous process
True ribs (1–7)	Attach to sternum by SEPARATE cartilaginous extensions (costal cartilage)
False ribs (8–10)	Share cartilaginous attachment (join cartilage of rib 7)
Floating ribs (11–12)	4 ribs total (2 pairs); NO sternal attachment at all
Spinal tap needle tip	Must reach SUBARACHNOID SPACE (below arachnoid, above pia)
Posterior = sensory	Posterior horn/root = sensory; anterior horn/root = motor (Bell-Magendie)
Gray commissure	Axons CROSSING OVER pass through here (bridges left/right gray matter)
Spinal shock	Transient/permanent loss of reflex activity; follows spinal cord injury
Pudendal nerve	Most active during sexual activity; S2-S4; also perineum sensation
Cervical plexus innervates	Levator scapulae, scalenes, sternocleidomastoid
Saphenous nerve	Largest cutaneous nerve; from LUMBAR portion of lumbosacral plexus
C8-T1 damage	Difficulty TYPING (intrinsic hand muscles; flexors of fingers)
Medial epicondyle	Compression → ulnar nerve injury ("funny bone")
Fibula fracture	Common fibular nerve damage → foot drop
Biceps/brachioradialis reflex	Tests C5–C7 segments
Reflex arc order	Receptor → Sensory (afferent) → Integration center → Motor (efferent) → Effector (1→2→3→4→5)
Monosynaptic reflex	Sensory neuron synapses DIRECTLY on motor neuron (no interneuron)
Higher centers & reflexes	Can ENHANCE or SUPPRESS spinal reflexes via descending tracts
Polysynaptic reflexes	More complex; interneurons activate MANY muscle groups simultaneously
Triceps jerk	Most appropriate to test for C7–C8 segment damage
Spinal meninges functions	Protect cord; maintain CSF flow; structural support (all correct)
Cervical plexus overview	All of the above = correct; multiple functions
Epidural near lower lumbar	Regional anesthesia; does NOT produce local anesthesia everywhere (FALSE for some claims)
Appendicular skeleton	Pectoral girdle + UE bones + pelvic girdle + LE bones